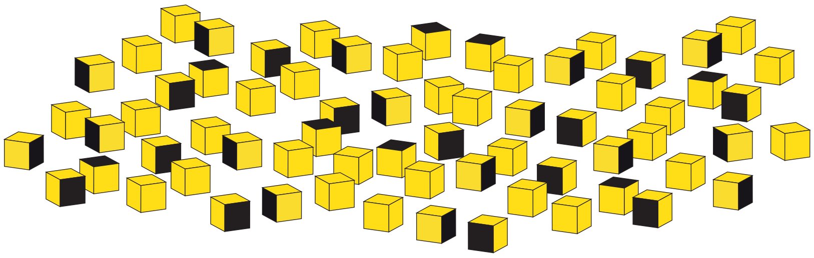
Examiner only

8. The atoms of radioactive substances decay at random.

(a) Students are given 200 cubes each with one side coloured black.
Describe how they would use the cubes to model radioactive decay **and** find the half-life. [6 QER]



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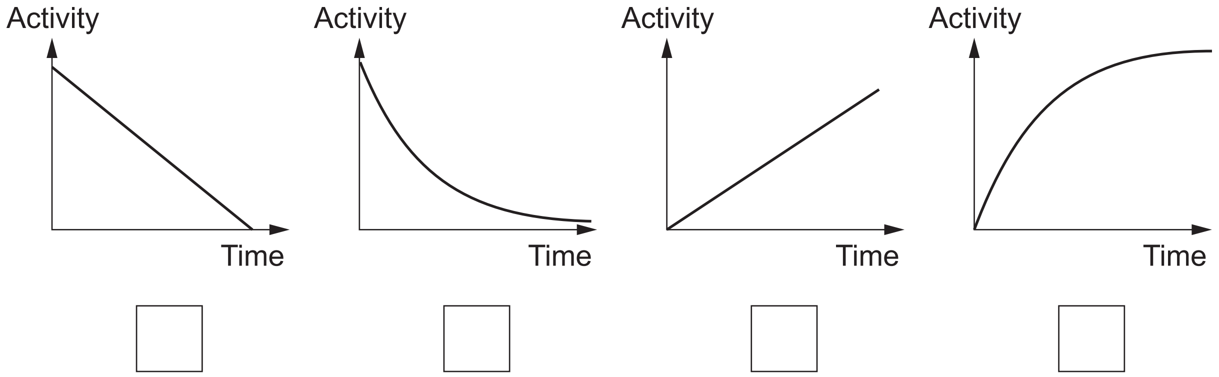
(b) An activity-time graph can be drawn for a radioactive substance.

Place a **tick (✓)** in the box beneath the graph that shows how the activity of a radioactive substance changes with time. [1]



(c) In an extension to their experiment, each group of students in a class is given 270 6-sided dice.
After each throw they remove the dice that show a ONE or a SIX facing upwards.

Their results are shown below.

Number of throws	Number of ONES removed	Number of SIXES removed	Number of dice remaining
0			270
1	40	....	185
2	37	30	118
3	18	17	....
4	14	14	55
5	10	9	36

(i) After the **first throw**, how many ONES would you have **expected** to be removed? [1]

number of ONES expected ....

(ii) **Complete the table.** [2]

(iii) Why does combining results from several groups of students better model radioactive decay? [1]

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